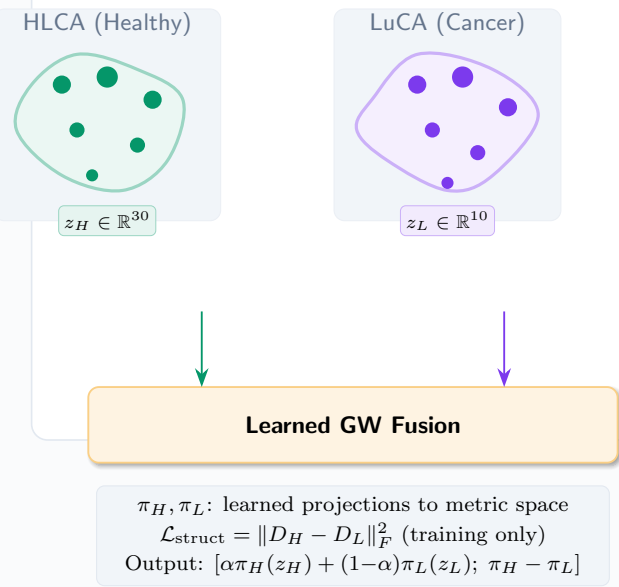
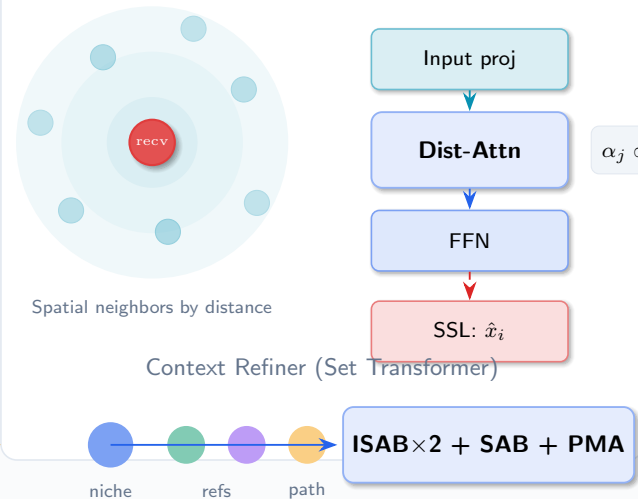


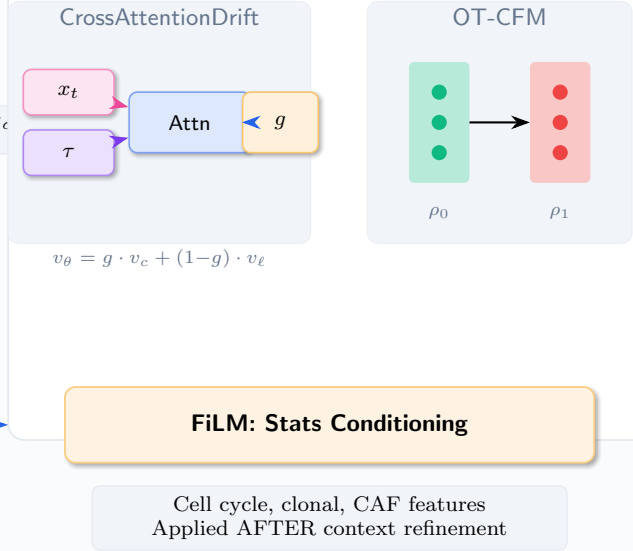
1. Reference Geometry



2. Niche Encoding (AMICI)



3. Transition Dynamics



Training Objectives



Core Equations

1. AMICI Distance-Weighted Attention

$$\alpha_{ij} = \frac{\exp(-d_{ij}/\sigma_h) \cdot \exp(q_i^\top k_j/\sqrt{d})}{\sum_{k \in \mathcal{N}_i} \exp(-d_{ik}/\sigma_h) \cdot \exp(q_i^\top k_k/\sqrt{d})}$$

Each head h learns its own distance scale σ_h

2. SSL Receiver Reconstruction

$$\hat{x}_i = f_\theta\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \cdot v_j\right), \quad \mathcal{L}_{\text{ssl}} = \|x_i - \hat{x}_i\|_2^2$$

Predict receiver from neighbor context only (no residuals)

3. Learned GW Fusion (Per-Cell)

$$h = \pi_H(z_H), \quad l = \pi_L(z_L), \quad z_f = [\alpha h + (1-\alpha)l; h - l]$$

$$\mathcal{L}_{\text{struct}} = \|D_H - D_L\|_F^2 \text{ where } D_H[i, j] = \|h_i - h_j\|$$

Structure loss guides training; inference is per-cell

4. Hierarchical Set Transformer

$$\mathbf{c} = \text{PMA}(\text{SAB}(\text{ISAB}(\text{ISAB}([e_{\text{niche}}, z_H, z_L, p]))))$$

FiLM: $\mathbf{c}' = \gamma(s) \odot \mathbf{c} + \beta(s)$ where s = stats token

Stats conditioning AFTER attention (prevents shortcut)

Transition Dynamics

CrossAttentionDrift (Gated)

$$v_c = \text{CrossAttn}(Q=[x_t, \tau], KV=\mathbf{c})$$

$$v_\ell = \text{MLP}([x_t, \tau])$$

$$v_\theta = g \odot v_c + (1-g) \odot v_\ell, \quad g = \sigma(W_g[\cdot])$$

OT-CFM Loss

$$(x_0, x_1) \sim \pi_{\text{Sinkhorn}}^*, \quad x_t = (1-t)x_0 + tx_1$$

Symbol	Dim	Description
x_i	$\mathbb{R}^{46}$	Cell embedding (HLCA 30 + LuCA 10)
d_{ij}	$\mathbb{R}^+$	Spatial distance (microns)
$\mathcal{N}_i$	set	K -nearest neighbors of cell i
z_H, z_L	30, 10	HLCA/LuCA reference latents
p	$\mathbb{R}^{14}$	PROGENy pathway activities
s	$\mathbb{R}^5$	Stats token (cycle, clonal, CAF)
$\mathbf{c}$	$\mathbb{R}^{256}$	Niche context embedding
v_θ	$\mathbb{R}^{40}$	Predicted velocity
τ	$\mathbb{R}^{32}$	Sinusoidal time embedding